

Madhusmita Talukdar

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Projects

- OpsPulse | [link](#) | Python, PostgreSQL, FastAPI, Kafka, Spark, XGBoost, Docker, Kubernetes, AWS** **Sept' 2026**
- Engineered a real-time data pipeline integrating PostgreSQL, Debezium CDC, Kafka AND Spark Structured Streaming with Apache Iceberg for organizing processed order and inventory data.
 - Implemented XGBoost demand forecasting and stockout-risk models with Isolation Forest anomaly detection, and integrated predictions into FastAPI REST services with a Streamlit dashboard.
 - Automated CI/CD and cloud deployment using Jenkins, GitHub, Kubernetes, Terraform, and AWS, with HPA-based scaling and Prometheus/Grafana monitoring.
- Reddit–Google Trends Engagement Analysis | [link](#) | Power BI, Data Modeling, DAX, Power Query** **Dec' 2025**
- Built an end-to-end analytics solution by extracting 9K+ Reddit posts through the Reddit API and integrating domain-level Google Trends data to analyze relationships between public search interest and community engagement.
 - Performed data preprocessing and transformation using Power Query, building structured data models and KPIs including engagement score, virality percentage, and demand-supply index for domain-level analysis.
 - Derived key insights, including sports contributing 56% of total virality, while identifying underserved domains through demand-supply gaps to support data-driven content strategy.
- Smart URL safety checker | [link](#) | ML Modeling, Scikit-learn, Pandas** **May' 2025**
- Engineered 30+ lexical and domain-based features from 70K+ URLs to build a phishing URL detection pipeline, enabling effective identification of malicious websites.
 - Trained and compared multiple models (Random Forest, XGBoost, Logistic Regression, Naive Bayes, and KNN), achieving 97% classification accuracy using hyperparameter tuning and evaluation with Precision, Recall, F1-score and Cross-Validation.
 - Deployed the best-performing model using Flask, creating a real-time web application for phishing URL classification with instant prediction capability.
- Los Angeles Crime Data Analysis | [link](#) | Python, Pandas, NumPy, Matplotlib, Seaborn** **Apr' 2025**
- Processed 2.5M+ Los Angeles crime records, performing data cleaning, preprocessing, feature engineering, and exploratory data analysis using Python, Pandas, and NumPy.
 - Identified temporal, geographic, demographic, and crime-type patterns using statistical analysis, heatmaps, and time-based visualizations to uncover crime trends and hotspots.
 - Analysed seasonal and COVID-period variations and crime clearance patterns to derive data-driven insights for understanding public safety trends.

Certificates / Certifications

- Supervised Machine Learning | Deeplearning.ai | [Certificate Link](#) **Apr' 2026**
- SQL (intermediate) certificate | HackerRank | [Certificate Link](#) **Oct' 2025**
- Human Computer Interaction | NPTEL | [Certificate Link](#) **May' 2025**

Skills

Programming: Python, SQL, C++

Data Analytics & ML: Pandas, NumPy, Scikit-learn, Statistical Analysis, EDA, Data Cleaning, Data Validation, Feature Engineering, Predictive Modelling, Hyperparameter Tuning

Data Engineering: Apache Spark, Spark Structured Streaming

Analytics & BI: Power BI, Microsoft Excel, PowerPoint, Power Query, DAX, Data Modelling, Data Visualization

Database & Backend: PostgreSQL, MySQL, SQL Server, FastAPI

Tools: Git, GitHub, Jupyter, Google Colab, VS Code, Jira

Others: Prompt Engineering, RAG, Agile Methodologies

Core Competencies: Analytical Thinking, Problem Solving, Communication, Teamwork, Adaptability

Education

- Lovely Professional University** **2023 – Present**
Bachelor of Technology - Computer Science and Engineering – CGPA: 8.07 *Phagwara, Punjab*
- Spectrum Gurukul** **2020 – 2021**
12th – Percentage: 92.6 *Nalbari, Assam*
- Sankardev Sishu Bidya Niketan** **2018 – 2019**
10th – Percentage: 92.8 *Nalbari, Assam*